Towards Personalized Precision Medicine Using DNA-Based Molecular Communication Networks

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ABSTRACT

One of the biggest problems in modern medicine is diagnosing deviations from an individual norm. Most diagnostic methods rely on comparison with a population average. However, this does not account for personal variations. DNA-based nanonetworks offer an alternative approach. This paper presents a novel DNA-based nanonetwork architecture that measures several individual health parameters and memorizes them. Based on that, the network determines deviations from that norm and communicates them to an external agent.

CCS CONCEPTS

• **Networks** → **Network architectures**; *Mobile networks*; *Short-range networks*; • **Hardware** → **Biology-related information processing**.

KEYWORDS

DNA, molecular communication, nanonetworks, nano communication, nano, nanomedicine

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1 PERSONALIZED NANONETWORKS

As a full scope introduction to DNA-based nanonetworks is beyond the scope of a short paper, we recommend reading [1] for the necessary definitions and mathematical preliminaries.

In classical medicine, medication is usually determined by comparison to a deviation from the population average, and the treatment follows a one-drug-fits-all model. While this can be a useful tool for the population average, it fails to account for individual variation.

To solve this problem of generalization, we conceptualize a nanonetwork that first measures normal values for an individual and later detects anomalies in comparison. Thus, we present a functional nanonetwork design to solve this general problem while also allowing for immediate treatments.

1.1 Phase 1

We divide the task (see Figure 1) of the nanonetwork into two main phases: first, measuring, and later monitoring and treatment. The entire process starts once all tiles and the seed-tile σ are present in the medium. Once started, Nanosensor^1–Nanosensor^n detect different health parameters and release specific tiles upon detection. This continues for a time t where message molecules of arbitrary length can form in part 3. Once time t elapsed, the orange tiles R, B, and M are introduced and stop the assembly process by adding a receptor to the message molecule that prevents further assembly. The thus assembled message represents the concentration k of a parameter in unary coding.

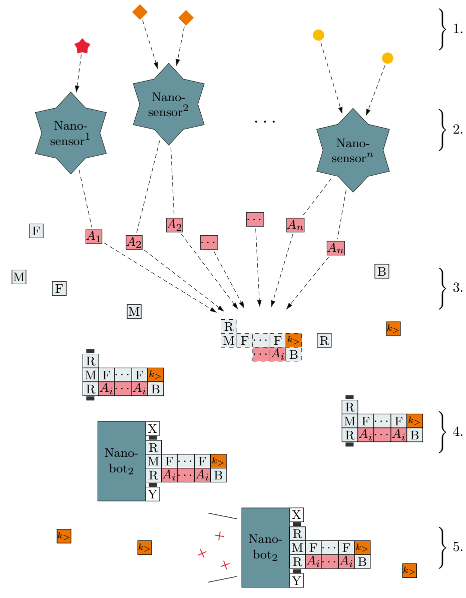

Figure 2: The nanonetwork tries to detect deviations from the normal range for the given health parameters.

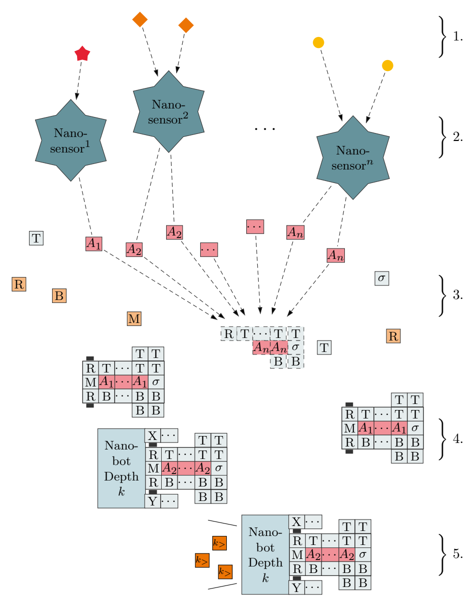

Figure 1: The nanonetwork computes the "normal range" of given health parameters.

The fully assembled messages can then bind to fitting boxes of maximum depth k like a key. Each of the opened boxes releases new seed tiles $k_>$ for phase 2. The boxes also release tiles that bind to all seed tiles $k - i_>$ and thus ensure the presence of only a single seed tile that represents a normal health parameter. Thus, the measured concentration is converted from unary coding to a single tile. Then the network enters phase 2.

1.2 Phase 2

Phase 2 is depicted in Figure 2. The initially released seed-tiles $k_>$ begin forming message frameworks of length k plus a predefined threshold j. For the same time interval t, the messages are now allowed to fill the assembled frame before the medical test expires. If the messages fully assemble in the given time, this signifies the reaching of a critical, predefined threshold. The finalized messages can then bind to a different type of nanobot that either immediately reacts by releasing medication or by communicating the measurements to the outside.

We used the ISUTAS software from Patitz [3] to evaluate the proposed assemblies. The molecules shown in Section 1 have been instantiated and tested in the corresponding modules for kTAM and HAM in [2].

The molecule assembles without growths/facet errors.

2 CONCLUSION

The proposed nanonetwork can measure a concentration value and persistently store the measured value as a tile. The network learns a "normal range" for a number of health parameters and can directly react upon exceeding a predefined range. This general approach is superior to regular medical tests, as both memory and computational power can be utilized for many disease markers and potentially assist with yet unsolved problems.

3 ACKNOWLEDGEMENT

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